

## Deterministic pose-spline dynamics estimator

Shared parameter update: nominal -> estimated

|                       |                            |                                   |
|-----------------------|----------------------------|-----------------------------------|
| mass [kg]             | 2.351597591 -> 2.889318105 | delta=+0.537721                   |
| ratio=1.2286618       |                            |                                   |
| CoG x [m]             | 0 -> -0.00554341987        | delta=-0.00554342 ratio=-         |
| CoG y [m]             | 0 -> -0.006856377694       | delta=-0.00685638 ratio=-         |
| CoG z [m]             | 0 -> 0.003996351346        | delta=+0.00399635 ratio=-         |
| rotor effectiveness 1 | 1 -> 1.070629397           | delta=+0.0706294 ratio=1.0706294  |
| rotor effectiveness 2 | 1 -> 1.033330018           | delta=+0.03333 ratio=1.03333      |
| rotor effectiveness 3 | 1 -> 0.9787111466          | delta=-0.0212889 ratio=0.97871115 |
| rotor effectiveness 4 | 1 -> 0.9235645521          | delta=-0.0764354 ratio=0.92356455 |
| command lag [s]       | 0.01 -> 0.07653898619      | delta=+0.066539 ratio=-           |

Inertia matrix [kg m<sup>2</sup>]

nominal:  
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]  
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]  
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]  
estimated:  
[ 0.0769459969, -0.001721323942, 7.250480827e-05]  
[-0.001721323942, 0.08248039315, 2.793502766e-05]  
[ 7.250480827e-05, 2.793502766e-05, 0.1584645146]

Principal moments: nominal -> estimated

|                                         |                                |                  |
|-----------------------------------------|--------------------------------|------------------|
| principal moment 1 [kg m <sup>2</sup> ] | 0.06495264516 -> 0.07645423544 | delta=+0.0115016 |
| ratio=1.1770765                         |                                |                  |
| principal moment 2 [kg m <sup>2</sup> ] | 0.06500006701 -> 0.08297208094 | delta=+0.017972  |
| ratio=1.2764922                         |                                |                  |
| principal moment 3 [kg m <sup>2</sup> ] | 0.1289921163 -> 0.1584645883   | delta=+0.0294725 |
| ratio=1.2284827                         |                                |                  |

Optimization

joint dynamics loss: 0.0305405510747  
soft-prior cost: 0.0178326331951  
selected strict-ZOH lag [s]: 0.076538986185

Bag success\_single

reconstruction position RMSE [m]: 0.000189532383255  
reconstruction position component RMSE [m]: [9.684228e-05 1.306442e-04 9.734567e-05]  
reconstruction orientation RMSE [deg]: 0.158250443634  
gyro consistency RMSE xyz [rad/s]: [0.013531 0.014215 0.009353]  
specific-force consistency RMSE xyz [m/s<sup>2</sup>]: [0.169057 0.0559 0.032697]  
inferred external force RMS xyz [N]: [0.253013 0.146988 0.221451]  
inferred external torque RMS xyz [N m]: [0.041533 0.059048 0.105171]  
bag dynamics loss: 0.0305405510747